

Data and Metadata

K. Gibert

*Knowledge Engineering and Machine Learning group at
Intelligent Data Science and Artificial Intelligence Research Center*

Universitat Politècnica de Catalunya, (IDEAI-UPC) Barcelona

Dean of Illustrious Professional College of Informatics Engineering of Catalonia

President and founder of donesCOEINF



<https://www.eio.upc.edu/en/homepages/karina>,



karina.gibert@upc.edu,



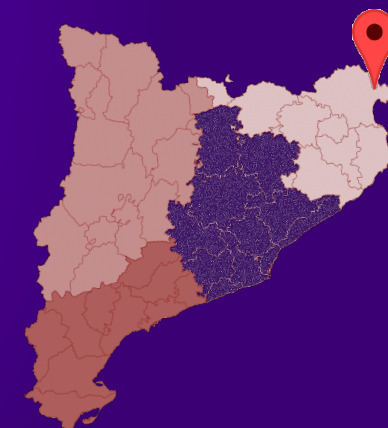
[karina.gibert](#),



[@karinagibertk](#)



Figuerenca



Planeta , Abril, 2024

© K. Gibert



Basic structure for analysis

The data matrix

	Weight	Height	Sex	Eyes
John	85	1.85	M	azul
	.	.	.	.
	.	.	.	.



Rows: Individuals (study units) ($i1...in$)

Columns: Variables (characteristics of individuals) ($X1..Xk$)

Cells: Value of variables for individuals (xik)

Type of variables

- Numerical: Quantitative, measure

Categorization

continuous (real quantity):

Weight, Height

Discretization

discrete (natural quantity):

Age, shoes size

- Categorical: Qualitative, adjective

(eventually codified)

Ordinal (ordering over modalities):

Socioeconomic status

Binary (two modalities):

wear glasses

Nominal (unordered modalities):

Hair color

RE
Categorization

- Date: Special formats, only some softwares

- Other variables

(no standard, rarely used in standard data mining applications)

- Fuzzy variables

- Hierarchical variables

- Multivalued variables

- Compositional variables

- Semantic variables

- Distributional variables

- Interval variables/Ratio variables (means, standard dev, dotplots)

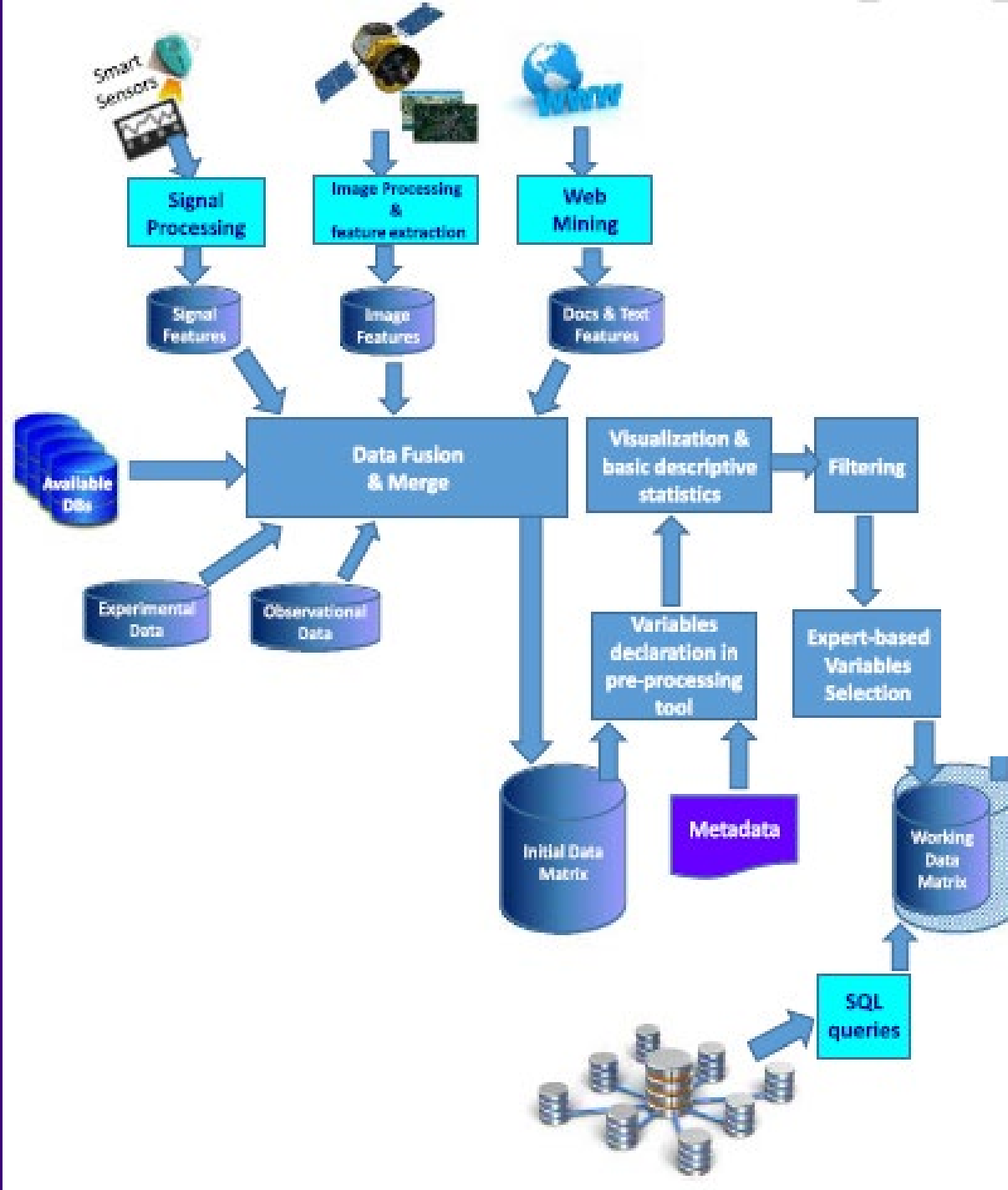
- Textual data

Loss
information

Better
avoid

Mean/StDev
Histograms

Percentages
Tables
BarPlots



Data Sources

Methodology

[Gibert Aicomm2016]

Gibert, K., M. Sánchez-Marrè, J. Izquierdo (2016) A Survey on Pre-processing Techniques in the Context of Environmental Data Mining. Artificial Intelligence in Communications, 29(6): 627-663, IOSPress DOI: 10.3233/AIC-160710

Type of data

- Primary/secondary
- Experimental/observational
- Open/Private
- Sample/Population

From Data to Decisional Knowledge

DATA  **INFORMATION**

prepostdif.DAT - WordPad

Home View

Clipboard: Paste, Cut, Copy

Font: Courier New, 11, Bold, Italic, Underline, Text color, Background color

Paragraph: Bulleted list, Numbered list, Decrease indent, Increase indent, Paragraph style

Insert: Picture, Paint drawing, Date and time, Insert object

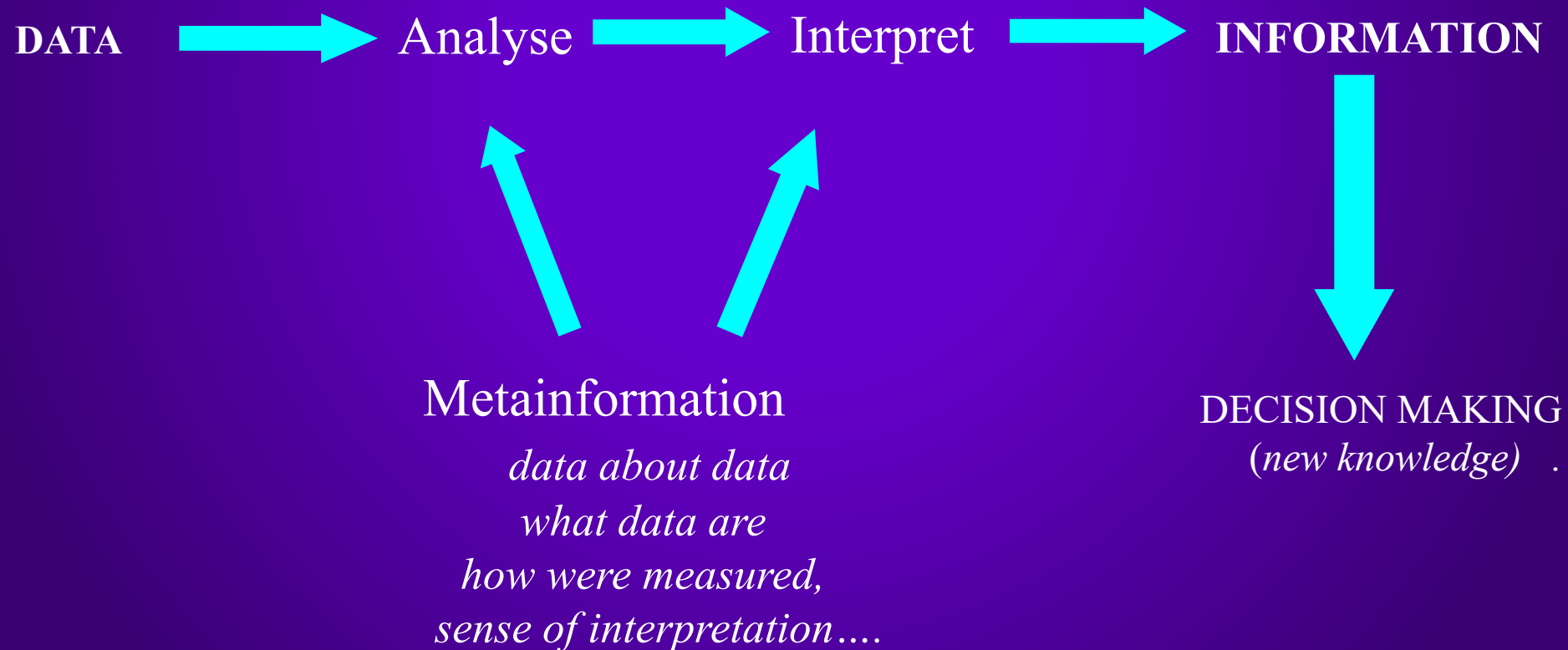
Editing: Find, Replace, Select all

3 2 1 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

((0	5	5	300	300	0	0	35	35	0	2	2
	0	0	0	0	1	1	?	?	?	500	
	500	0	-25	-4	21	0	0	0	50	50	0
	36	0	-36)								
(((6	6	0	300	300	0	0	0	0	3	3	0
	0	38	38	0	0	0	?	?	?	500	
	500	0	-25	-25	0	0	0	0	50	50	0
	24	30	6)								
(((5	5	0	300	78	-222	36	40	4	5	5	0
	0	0	0	0	0	0	?	?	?	500	
	200	-300	1.72	3.24	1.52	0	6	6	50	18	-
32	21	42	21)								
(((6	6	0	300	33	-267	0	35	35	4	4	0
	41	47	6	1	3	2	?	?	?	500	80
	-420	-25	-8.75	16.25	0	5	5	50	26	-24	39
	60	21)									
(((7	6	-1	82	52	-30	40	44	4	2	4	2
	38	53	15	0	6	6	?	?	?	340	
	183	-157	15.09	8.31	-6.78	2	5	3	43	28	-
15	39	39	0)								
(((0	5	5	300	100	-200	0	30	30	0	3	3
	0	54	54	0	6	6	?	?	?	500	
	210	-290	-25	5	30	0	4	4	50	20	-
30	30	0	-30)								
(((0	0	0	300	300	0	0	0	0	0	0	0
	0	0	0	0	0	0	?	?	?	500	
	500	0	-25	-25	0	0	0	0	50	50	0
	0	0	0)								
(((6	5	-1	60	120	60	11	15	4	4	4	0
	55	53	-2	10	6	-4	?	?	?	300	
	220	-80	-0.11	2.49	2.6	6	6	0	7	9	2

From Data to Decisional Knowledge

DATA  **INFORMATION**



Metadata

Data Origin: Secondary source/Primary

Inclusion criteria: Representativity? Target?

Size of data: $n \times K$ ($n > 10K$)

All variables:

- What is it measuring (Measuring tool or procedure)
- Measuring unit
- Representation of missing data
- Meaning of variable

Quantitative variables:

- Range of possible values

Qualitative variables:

- Set of possible modalities
- Representation of modalities
- Meaning of modalities

Role of variables: Response/Explanatory

Still an open
problem

Software do not support

- External project documentation manually managed

- Relational Data Base for very complex Data Matrices [Gibert, MMR 92]

Gibert, K., & Marti-Recober, M. (1992). A System for Production and Analysis of Statistical Reports. In *Computational Statistics* (pp. 363-368). Physica-Verlag HD.

Metadata File

url: www.xxx.ssss.www

Inclusion criteria: *People in [18,65] years, no hard attacks, no smoking, no cholesterol, married, with sons or daughters....*

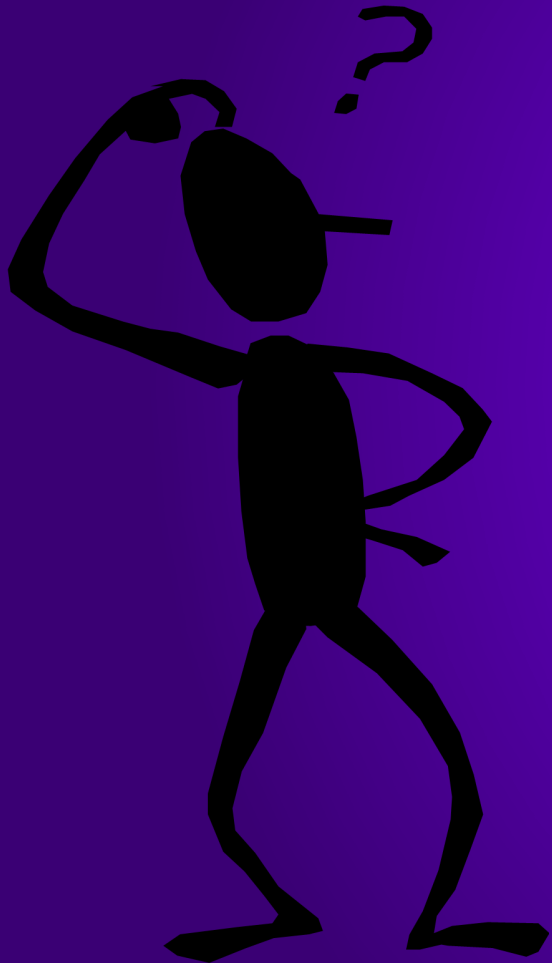
n: *nro of rows*

K: *nro of columns*

Variable	Short Variable Name	Modalities	Short Mod Name	meaning	Type	Measuring unit	Missing code	Measuring procedure	Range	Role
Age	Age			Age of marriage	Num	years	""		[1,105]	Explanatory
Sex	Sex			Biological sex	Quali		Unk			Explanatory
		M		Male						
		H		Female						
Concentration of Iron	FeC			Level of Iron in blood	Num	µg/dl	NA	Biochemical analysis on blood sample measuring transferrine.....	[30, 200]	Explanatory
Has anemiy?	Anemiy			person anemiy diagnosis	Boolean		Unknown	Levels of Fec<xxx and		Response
Date of birth	Date			Date of birth of the person	DATE	Yyy:mm:dd	NA			Explanatory

First insight to Data

- Look at Metadata
- Determine rows and columns to be kept for the analysis
- Basic descriptive analysis of remaining variables
 - Inspect anomalies, errors, missing data, outliers
- First report about data quality
- Preprocessing
- Verify after each processing step
- Final descriptive analysis (*report data improvements*)



Any question?...

K. Gibert

<https://www.eio.upc.edu/en/homepages/karina>
karina.gibert@upc.edu  @Karinagibertk

*KEMLG-@-IDEAI: Knowledge Engineering and Machine Learning group at
Intelligent Data Science and Artificial Intelligence Research Center*

Universitat Politècnica de Catalunya, Barcelona

*Dean of Illustrious Professional College of Informatics Engineering
of Cataloonia (COEINF)*

Founder and president of donesCOEINF